

Rolling bearing fault diagnosis method based on MTF-MMCNN

Ruicheng Feng, Qiyue Zhang, Lu Wang, Manwen Li & Chunli Lei

To cite this article: Ruicheng Feng, Qiyue Zhang, Lu Wang, Manwen Li & Chunli Lei (2026) Rolling bearing fault diagnosis method based on MTF-MMCNN, Nondestructive Testing and Evaluation, 41:2, 761-781, DOI: [10.1080/10589759.2025.2474102](https://doi.org/10.1080/10589759.2025.2474102)

To link to this article: <https://doi.org/10.1080/10589759.2025.2474102>



Published online: 01 Mar 2025.



Submit your article to this journal [↗](#)



Article views: 387



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 18 View citing articles [↗](#)



Rolling bearing fault diagnosis method based on MTF-MMCNN

Ruicheng Feng, Qiyue Zhang , Lu Wang, Manwen Li and Chunli Lei

School of Mechanical and Electronical Engineering, Lanzhou University of Technology, Lanzhou, China

ABSTRACT

During the operation of bearing, complex and variable working conditions and insufficient samples will lead to low accuracy of bearing fault classification using traditional methods. To solve these problems, a mixed multi-scale convolutional neural network (MMCNN) fault diagnosis model combined with Markov transition field (MTF) method is proposed. Firstly, the MTF is utilised to convert the one-dimensional vibration signal of the bearing into a two-dimensional image with more sufficient time information. Then, the multi-scale branch module and the multi-scale cascade module are fused to fully extract the image features, a Convolutional Fusion Attention Module (CFAM) is constructed to strengthen the important features and assign appropriate weight to each channel, and a dropout layer is designed to make the model more stable. Finally, the images are input into the MTF-MMCNN to achieve the classification of bearing fault, and the performance of the model is verified by using the CWRU bearing dataset and the MFS bearing dataset in our laboratory. The results show that MTF-MMCNN has higher classification accuracy and stronger robustness under different working conditions, which is beneficial for formulating maintenance strategies.

ARTICLE HISTORY

Received 27 November 2024

Accepted 24 February 2025

KEYWORDS

Fault diagnosis; Markov transition field; mixed multi-scale convolutional neural network; convolutional fusion attention module; rolling bearing

1. Introduction

With the continuous promotion of the ‘Intelligent Manufacturing’ strategy, modern mechanical equipment has developed towards automation, efficiency and intelligence [1]. Rolling bearing is one of the important parts of mechanical system [2], playing an indispensable role in fields such as automobile [3], aviation [4] and wind power [5]. Its performance will affect the working efficiency of mechanical equipment [6]. Due to the harsh working environment, rolling bearings are prone to faults such as pitting, cracks and spalling at the inner and outer ring and rolling element positions during operation. Data show that approximately 40–44% of equipment failures are related to bearings [7]. The bearing vibration signal contains bearing fault information, which can be analysed and processed to identify the health status of the bearing [8]. In the early stage of actual operation, the fault characteristics of bearing signals are not obvious, resulting in a small sample size and difficulty in obtaining fault signals. At the same time, rolling bearings

CONTACT Chunli Lei  lclyq2004@163.com

This article has been corrected with minor changes. These changes do not impact the academic content of the article.

© 2025 Informa UK Limited, trading as Taylor & Francis Group

usually operate under complex and changeable working conditions such as variable load and speed. Therefore, it is of great significance to extract bearing fault signal characteristics and diagnose fault categories under conditions of variable working and insufficient samples.

Traditional rolling bearing fault diagnosis methods require manual signal processing, relying on signal processing technology and expert experience [9]. Although the traditional fault diagnosis method can effectively predict the bearing fault, it is subject to more human intervention, and there are often problems such as poor robustness, weak generalisation performance, and difficulty for technical personnel to perform fault diagnosis tasks [10]. Hinton et al. [11] first proposed the concept of deep learning, which mines more valuable feature information in data by establishing a machine learning model with multiple hidden layers. Deep learning is widely used in image recognition [12], natural language processing [13], semantic segmentation [14] and speech processing [15], and is now also involved in the field of fault diagnosis.

With the development of computer hardware, the current research on bearing fault diagnosis is mostly based on deep learning, among which Convolutional Neural Network (CNN) is a popular method [16,17]. Its powerful data mining ability and feature adaptive learning ability have attracted great attention [18–20]. However, researchers have gradually dissatisfied with the practical application of CNN, and more people have improved it. Feng et al [21] presented a 1DCNN model based on batch normalisation (BN) layer for intelligent fault diagnosis of rotating machinery, and CNN was adopted for one-dimensional vibration signal analysis to finally realise fault classification. Liu et al. [22] put forward a multi-task one-dimensional convolutional neural network (MT-1DCNN), and utilised the features learned by related tasks to improve the performance of fault diagnosis model. Aiming at the lack of original input signal information, Huang et al. [23] proposed a multi-scale convolutional neural network integrating different scale filters to extract more sufficient bearing fault feature and improve the fault diagnosis effect. Although the above improvement of CNN network structure can enhance the training efficiency, there is still a problem of loss of fault information of original signal in the field of bearing fault diagnosis. How to effectively use the information in bearing signal has become a further research direction. Gu et al. [24] used Gram Angle Field (GAF) to process the original vibration signal and adopted Ghost module to improve ResNet model, and the effectiveness of GAF coding method was verified by comparative experiments. Wu et al. [25] employed one-dimensional vibration signal stacking to generate two-dimensional fault feature images, and combined with LeNet-5 with more convolutional pooling layers for fault diagnosis. Xue et al. [26] combined the two-dimensional coding method of Markov transition field (MTF) with CNN, and used multi-dimensional convolutional neural network to extract image features, which eliminated the labour cost of extracting fault information from bearing original signals in the past.

The above MTF method can retain more valuable information in the original vibration signal, but the feature extraction model constructed by CNN is obviously not enough to fully extract the fault features. With more in-depth research, the drawbacks of neural networks that cannot highlight key information have begun to emerge. The technology of simulating the attention mechanism of the human visual system, which can selectively focus on some key information and ignore other

redundant information, has gradually become a development direction for improving deep neural networks [27]. Hu et al. [28] proposed the Squeeze-and-Excitation Network (SENet), which included an attention structure for network channels that can redistribute weights to each channel and suppress irrelevant information, effectively improving the generalisation performance of the network model. Wang et al. [29] presented a data-driven intelligent fault diagnosis method based on multi-head attention and convolutional neural network. Xu et al. [30] put forward a fault diagnosis method based on Convolutional Block Attention Module (CBAM) to enhance InceptionNet, achieving robust feature extraction and accurate fault diagnosis. Although the above fault diagnosis methods have achieved high accuracy, many of them still have following shortcomings:

- (1) Many two-dimensional image conversion methods in bearing fault diagnosis are difficult to consider the time correlation of vibration signals, and it is easy to lose coding information.
- (2) In order to improve the accuracy of fault diagnosis, some models continue to superimpose the number of neural network layers. However, this approach may not only lead to an increase in model training parameters but also easily cause model overfitting.
- (3) The above research models can effectively classify bearing faults, but changes of load, speed and other working conditions during the actual operation of mechanical equipment will affect the accuracy of fault diagnosis, and the insufficient sample size will also restrict the training efficiency of the network model.

Therefore, in order to solve the above problems, this study proposes a rolling bearing fault diagnosis method based on MTF-Mixed Multi-scale Convolutional Neural Network (MTF-MMCNN). The main contributions of this paper are as follows:

- (1) The MTF coding method is introduced to convert the original one-dimensional vibration signal into a two-dimensional image, which retains the time correlation of the original vibration signal.
- (2) A mixed multi-scale module is presented combining the advantages of sufficient multi-scale branch feature extraction and high multi-scale cascade training efficiency to extract deep image features. Meanwhile, a Convolutional Fusion Attention Module (CFAM) is designed to assign corresponding weights to each channel, effectively improving the generalisation performance of the model.
- (3) A fault diagnosis model of MTF-MMCNN is constructed to effectively deal with the problem of low fault classification accuracy under the condition of many changes in the actual working conditions of rolling bearings and insufficient sample size.

The subsequent arrangement is as follows: [Section 2](#) introduces the basic theory, including MTF and the basic principle of CNN. In [Section 3](#) the CFAM module, MMCNN model and MTF-MMCNN fault diagnosis process are developed. [Section 4](#) is

experimental validation and analysis. [Section 5](#) is the comparative analysis experiment. [Section 6](#) summarises the conclusion of this paper.

2. Basic theory

In this section, the MTF is first introduced and then a detailed description of the CNN is provided.

2.1. MTF

In this paper, the MTF coding method is adopted to convert the one-dimensional vibration signal of the bearing into a two-dimensional feature image, which is input into the neural network for feature extraction. This method is based on the first-order Markov chain considering the time position relationship, which can retain the time correlation of the original one-dimensional data. The two-dimensional feature map transformed by MTF has richer fault information.

Assuming that the original one-dimensional data length is $n(X = \{x_1, x_2, \dots, x_n\})$, it is divided into Q quantile units, and each data point x_i is mapped onto $q_i (q_i \in \{1, 2, \dots, Q\})$. Thus, the Markov transition matrix W of $Q \times Q$ can be constructed, as shown in Equations (1) and (2).

$$W = \begin{bmatrix} w_{11} & w_{12} & L & w_{1Q} \\ w_{21} & w_{22} & L & w_{2Q} \\ M & M & O & M \\ w_{Q1} & w_{Q2} & L & w_{QQ} \end{bmatrix} \quad (1)$$

$$w_{ij} = P\{x_{t+1} \in q_i | x_t \in q_j\} \quad (2)$$

where w_{ij} is the probability that the quantiles q_i and q_j are adjacent, t is the time series, x_t is the amplitude of the bearing signal at time t , q_i is the quantile corresponding to the time series.

Since the Markov transition matrix W is insensitive to the distribution of the time series and loses the time information, the MTF coding method is introduced to improve the time correlation in the Markov transition matrix. The matrix M of $Q \times Q$ is constructed, as shown in Equations (3) and (4).

$$M = \begin{bmatrix} m_{11} & m_{12} & L & m_{1Q} \\ m_{21} & m_{22} & L & m_{2Q} \\ M & M & O & M \\ m_{Q1} & m_{Q2} & L & m_{QQ} \end{bmatrix} \quad (3)$$

$$m_{ij} = P(q_i \rightarrow q_j) \quad (4)$$

where m_{ij} is the probability that quantile q_i transfers to quantile q_j .

2.2. CNN

2.2.1. Convolutional layer

The convolutional layer and the pooling layer are the main components of the convolutional neural network. The convolutional layer is used to extract the hidden features of the image, and the convolutional kernel is utilised to scan the different positions of the feature image. The expression of the convolutional operation is shown as follows.

$$X_j^l = f \left(\sum_{i=1}^M \left(X_j^{l-1} * \omega_{ij}^l \right) + b_j^l \right) \quad (5)$$

where X_j^l is the output feature map of layer l , $f(\cdot)$ is the activation function, M is the number of output feature maps, ω_{ij}^l is the weight of the convolutional kernel, b_j^l is the bias of the layer l .

2.2.2. Pooling layer

The pooling layer is usually connected to the convolutional layer. The main function is to reduce the dimension of the feature map output by the convolutional layer and the computational complexity of the network parameters. The common pooling layer includes maximum pooling and average pooling, which means that the maximum or average value is used to replace the whole area information in a certain area of the image, so the pooling operation can be understood as the blur processing of the image. The formula is described as:

$$P_i^{l+1}(j) = \max_{(j-1)W+1 \leq t \leq jW} \{q_i^l(t)\} \quad (6)$$

where $q_i^l(t)$ is the output value of the t neuron in the i channel of the l layer, W is the size of the pooled nucleus, and $P_i^{l+1}(j)$ is the output value of the $l+1$ channel in the i layer.

2.2.3. Activation function and fully connected layer

The activation function is mainly employed to provide nonlinear modelling capabilities for the network model. A nonlinear activation function can improve the stability of neural network model training and accelerate model convergence. In this paper, the ReLU activation function is selected, and the expression is as follows.

$$f(x) = \begin{cases} x, & x \geq 0 \\ 0 & x < 0 \end{cases} \quad (7)$$

The fully connected layer is usually linked to the last layer of the network structure, which is used to map the fault features learned by the model to the sample space for classification. The fully connected layer converts the convolutional or pooling output of the previous layer into a one-dimensional feature vector, and then fully connects with the next layer. The fully connected layer operation is shown in Equation (8).

$$y^l = f \left\{ (w^l)^T x^{l-1} + b^l \right\} \quad (8)$$

where x^{l-1} is the output value of layer $l-1$, w^l is the weight, y^l is the output of layer l , b^l is the bias term.

The Softmax classifier belongs to a kind of activation function, which can ensure the non-negativity of the prediction probability. Normalisation can take the characteristics output by the full connection layer as input and transform them into a probability distribution with sum of 1. The fault category with the highest probability is the prediction result. The Softmax function formula is as follows:

$$P_i = \frac{e^{\theta_i^T x}}{\sum_{j=1}^K e^{\theta_j^T x}} \quad (9)$$

where $i = 1, 2, \dots, k$, K is the number of output nodes, P_i denotes the probability of each output, $\sum_{i=1}^K P_i = 1$.

3. Proposed method

Firstly, CFAM attention module and MMCNN structure are proposed. Secondly, based on the above MTF data processing method, the MTF-MMCNN fault diagnosis model is constructed. This section mainly introduces the above developed theories in detail.

3.1. CFAM attention module

The attention mechanism aims to assign different attention weights to different parts of the model input, and adjust the learning focus of the model on key information, thereby improving the performance of the model. SENet is the most common channel attention mechanism, but it uses a pooling method to compress input features into one dimension, which can easily lose fault information. Therefore, this paper proposes a convolutional fusion attention module (CFAM). Different from the SENet, CFAM utilises global average pooling (GAP) and global maximum pooling (GMP) to compress features to prevent partial information loss. One-dimensional convolutional operation replaces multi-layer perceptron to allocate attention weight more efficiently. The information of two branches is interacted and fused, and the channel features are recalibrated as a whole. The CFAM attention structure is shown in Figure 1.

CFAM includes feature compression and weight generation. In the feature compression stage, GAP and GMP are used to compress the input features along the high and wide dimensions, and the outputs are P_{GAP} and P_{GMP} respectively. In the weight

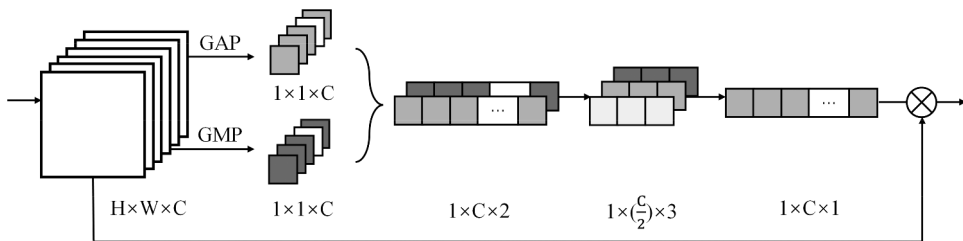


Figure 1. Convolutional fusion attention module (CFAM).

generation stage, the compressed features are tiled into a one-dimensional tensor and stacked along the channel dimension to facilitate the integration of one-dimensional convolutional operations in the attention mechanism module and output double-pooled features. One-dimensional convolution is adopted to reduce the size and the number of rising channels, and then the transposed convolution is selected to restore the size and the number of channels becomes 1. Finally, the sigmoid activation function is chosen to generate weights and multiply them with the input features. The whole CFAM calculation process is shown in Equations (10) to (14):

$$P_{GAP} = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W x(i, j) \quad (10)$$

$$P_{GMP} = \max x(i, j) \quad (11)$$

$$Y_p = \text{flatten}(P_{GAP}) \oplus \text{flatten}(P_{GMP}) \quad (12)$$

$$Y_T = \text{CovT}(\text{Cov}(Y_p)) \quad (13)$$

$$z = \text{reshape}(Y_T) \quad (14)$$

where P_{GAP} and P_{GMP} are one-dimensional data after GAP and GMP compression, respectively, i and j are the position of the input value x in the two-dimensional data, $\text{flatten}(\cdot)$ is a flattening operation, $\oplus$ denotes stacking along channel dimensions, Y_p is the pooled information after fusion, $\text{Cov}(\cdot)$ and $\text{CovT}(\cdot)$ represent convolutional and transposed convolutional operations, Y_T is the middle layer of weight information, $\text{reshape}(\cdot)$ represents the shape of $1 \times 1 \times C$, z is the final generated channel attention weight.

3.2. MMCNN

In the convolutional neural network, layers with different convolutional kernel sizes will form different receptive fields. The multi-scale convolutional neural network fuses the results of a series of convolutional operations with various convolutional kernel sizes. The output containing multiple scales makes the network feature extraction more sufficient. There are two main types of traditional multi-scale feature fusion networks. The first is a multi-scale branch structure, as shown in Figure 2, where AP represents

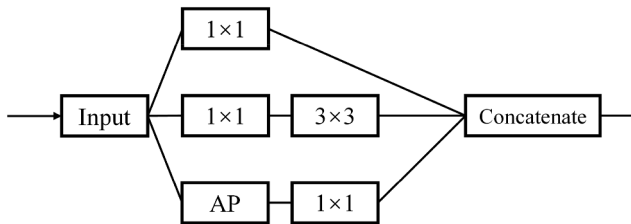


Figure 2. Multi-scale branch structure.

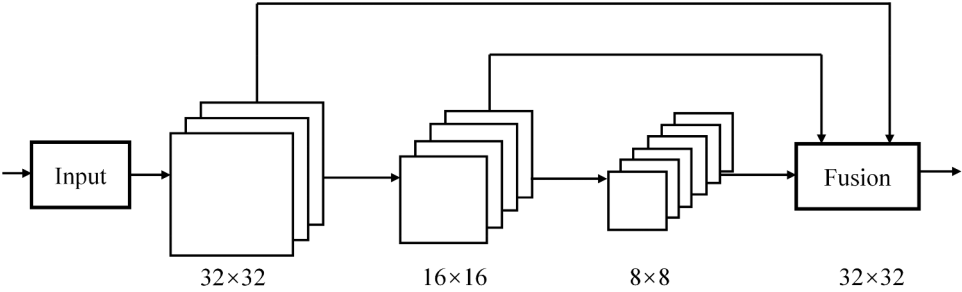


Figure 3. Multi-scale cascaded fusion network.

average pooling. The multi-scale structure is characterised by different convolutional kernels for each branch in the same layer of network, and 1×1 convolution is used to reduce the number of channels for simple operation. The second is a multi-scale cascade structure, as shown in Figure 3. There is only one backbone, the entire network chain gradually deepens from the shallow layer whereas the size of the feature map decreases, and the image obtained by up-sampling the small-scale features is fused with the large-scale feature map.

The multi-scale branch structure model has a small structure and cannot achieve high-precision fault classification when a small number of modules are stacked. Therefore, this paper presents MMCNN that integrates multi-scale branch structure and multi-scale cascade structure to avoid model over-fitting and long training time caused by complex network structure. At the same time, a BN layer is added to accelerate the convergence

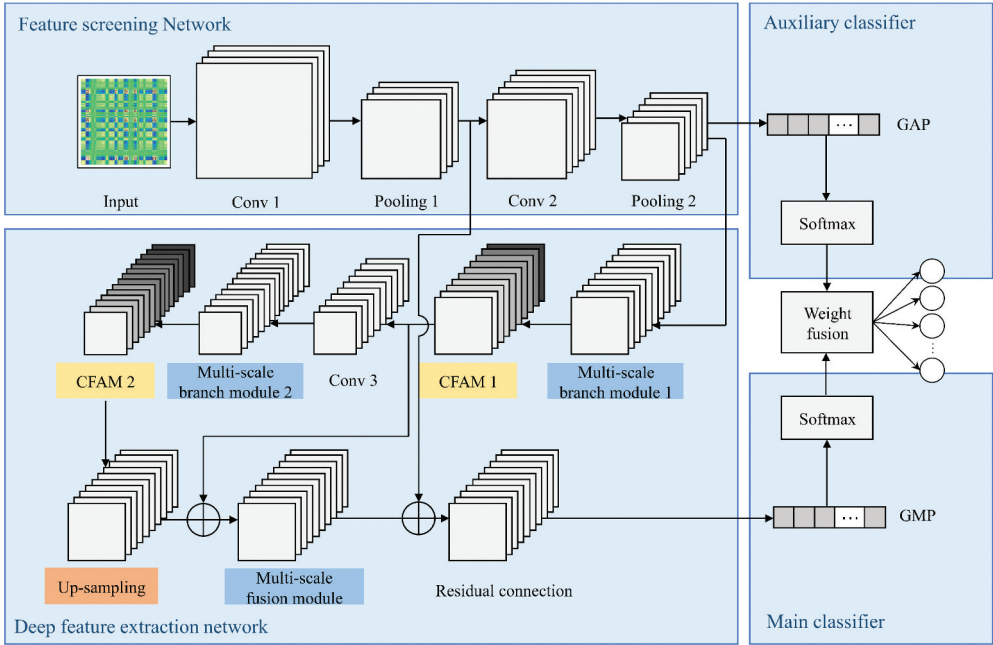


Figure 4. MTF-MMCNN structure.

speed of the model to prevent the gradient disappearance, and CFAM attention mechanism is introduced to improve the model's attention to different channel features. The MTF-MMCNN structure is shown in Figure 4, which includes four parts: (1) Feature screening network. (2) Deep feature extraction network. (3) Auxiliary classifier. (4) Main classifier.

The feature screening network is utilised to extract the initial features of the input image, including two convolutions and pooling operations. The first convolution applies a 5×5 large convolutional kernel to increase the receptive field of the network layer. After each pooling layer, a BN layer is applied to prevent overfitting. After preliminary feature extraction, the information is input into a deep feature extraction network to extract and fuse different scale features. The structure includes two multi-scale branch modules and a multi-scale cascade fusion module. The multi-scale branch module structure is shown in Figure 2, a total of three branches, among which 1×1 convolution is used to reduce the number of channels in order to aggregate information, and each branch information is stacked along the channel dimension for multi-scale feature fusion. Since each multi-scale branch module generates a large number of channels, the CFAM channel attention module shown in Figure 1 is introduced to adaptively assign weights to each channel information to suppress useless information. After the first multi-scale branch module, the down-sampling convolutional is adopted to reduce the size of the feature map. The small-scale features output by the second CFAM are restored to the image size through deconvolution and fused with large-scale features. The residual link is introduced to integrate information in the feature screening layer into deep feature to prevent information loss caused by the network depth. The main classifier chooses GAP instead of traditional Flatten layer and Dense layer to reduce computational complexity and accelerate the convergence of the model, and connects Softmax to classify faults. In addition, an auxiliary classifier composed of GAP and Softmax is added after the initial feature extraction. Finally, the main and auxiliary classifier complete the fault classification by weighted fusion.

The fault diagnosis process of MTF-MMCNN is shown in Figure 5. It mainly includes three stages. In the data re-processing stage, MTF is used to convert one-dimensional vibration signals into images. Then, the images are input into the MMCNN fault diagnosis model to output the classification results. Finally, the fault diagnosis results are displayed through model comparison and visualisation.

4. Experimental validation and analysis

The software environment for the experiments in this paper is the TensorFlow framework. The computer hardware configuration is Intel(R) Xeon-(R) Silver 4110 CPU, 64-GB RAM, NVIDIA Quadro P4000 GPU.

4.1. Data description

In this experiment, the bearing fault data of CWRU (Case Western Reserve University) bearing test data centre [31] and the MFS bearing test bench dataset of our laboratory were selected.

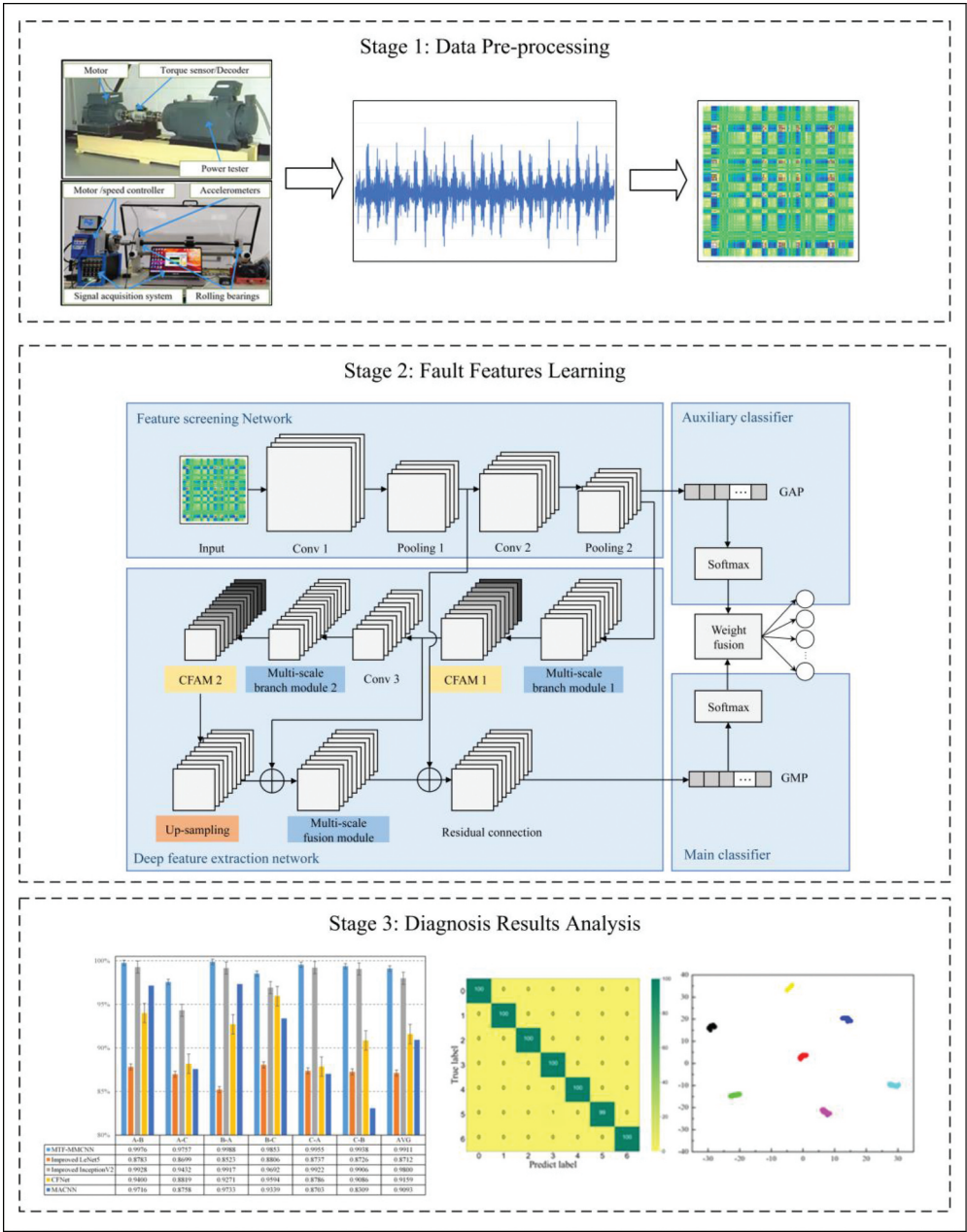


Figure 5. Diagnostic process of MTF-MMCNN.

The CWRU data set is from the bearing data acquisition test bench of Case Western Reserve University in the United States. The test object is the deep groove ball bearing of the motor drive end, and the bearing model is SKF6205. Each type of fault bearing is installed in the same state to test the vertical vibration signal. The CWRU bearing test rig mainly includes three-phase asynchronous motor, torque sensor and load, as shown in Figure 6. In this paper, three loads of 1hp, 2hp, and 3hp (1hp = 0.7457 kW) are selected to simulate three working

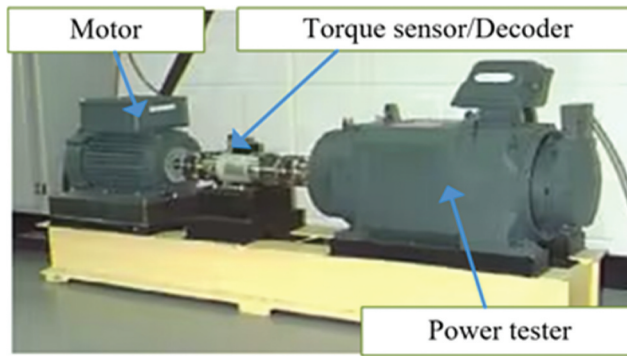


Figure 6. CWRU bearing test rig.

Table 1. Description of CWRU rolling bearing dataset.

		0	0.18 mm			0.36 mm			Rotational speed/r·mm ⁻¹
Fault diameter/mm		None	Inner race	Outer race	Ball	Inner race	Outer race	Ball	
Labels		0	1	2	3	4	5	6	
A	Training set	300	300	300	300	300	300	300	1hp(0.7457 kW)
	Test set	100	100	100	100	100	100	100	
B	Training set	300	300	300	300	300	300	300	2hp(1.4914 kW)
	Test set	100	100	100	100	100	100	100	
C	Training set	300	300	300	300	300	300	300	3hp(2.2371 kW)
	Test set	100	100	100	100	100	100	100	

conditions which are represented by labels A, B, and C. Bearing faults include three fault location types: outer ring, inner ring, and rolling element. Each fault type is divided into two diameter fault degrees of 0.18 mm and 0.36 mm. Therefore, there are 7 types of bearing faults including normal state. The signal sampling frequency is 12 kHz, and the minimum speed of the bearing is 1772 r/min. The data is segmented according to the length of the set sample. In this experiment, the length of the segmented sample is 2048, a total of 2800 samples, which are converted into 2800 initial feature maps by MTF. The 400 samples of each fault type are divided into training set and test set according to 3:1. The results are shown in [Table 1](#).

The MFS data set is derived from our test rig of Machinery Fault Simulator. A multi-channel SpectraPad data acquisition device is used in the experiment, and the signal processing and analysis are carried out in VQPro software. The test bearing is deep groove ball bearing ER-16k. The vibration signals of test bearing are transmitted by the AC collector to computer and the acquisition time is 6s. The MFS bearing test rig is shown in [Figure 7](#), which mainly includes fault bearing, healthy bearing, signal collector, signal acquisition system, motor and governor. The radial load added to the spindle between the two bearings is 50 N, and the sampling frequency is 15.36 hz. The preprocessing method of MFS dataset is same as that of CWRU dataset. The results are shown in [Table 2](#). The bearing faults includes three positions of inner ring, outer ring, and rolling body. The artificial processing fault damage degree is 0.6 mm and 1.2 mm. The dataset labels D, E and F represent the operation of the speed of 1200 r/min, 1300 r/min and 1400 r/min, respectively.

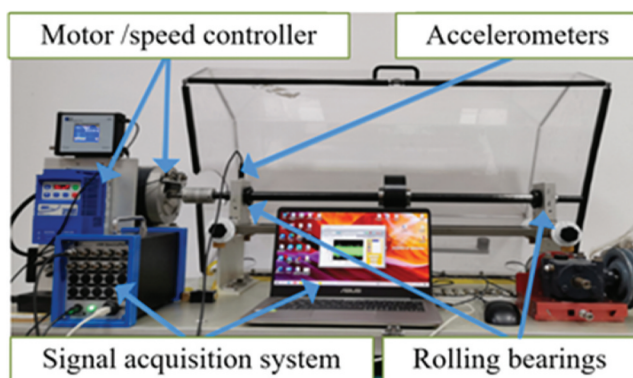


Figure 7. MFS bearing test rig.

Table 2. Description of MFS rolling bearing dataset.

		0	0.6 mm			1.2 mm			Rotational speed/r-mm ⁻¹
Fault diameter/mm		None	Inner race	Outer race	Ball	Inner race	Outer race	Ball	
Labels		0	1	2	3	4	5	6	
D	Training set	300	300	300	300	300	300	300	1200
	Test set	100	100	100	100	100	100	100	
E	Training set	300	300	300	300	300	300	300	1300
	Test set	100	100	100	100	100	100	100	
F	Training set	300	300	300	300	300	300	300	1400
	Test set	100	100	100	100	100	100	100	

4.2. Data preprocessing

The one-dimensional original data of 7 bearing fault types in CWRU dataset are sampled to obtain data samples, and then the two-dimensional images of 7 kinds of faults are obtained by MTF coding method, as shown in Figure 8.

4.3. Optimal parameter selection

In order to find the optimal model parameters, the training batch, learning rate, auxiliary classification weight and initial convolutional kernel size of the network model are adjusted respectively, and the training results are observed to select the parameters with the highest prediction accuracy. The test results are shown in Figure 9.

- ① **Batch size:** The batch sizes 4, 8, 10, 12, 24 are selected to test the performance of MMCNN model. It can be seen from Figure 9(a) that the accuracy of the model is the highest when the batch size is 12, so the batch size is set 12.
- ② **Learning rate:** The learning rates 0.05, 0.005, 0.0005 and 0.00005 are selected to test the performance of the model. In Figure 9(b), it can be seen that when the learning rate $\alpha \leq 0.005$, the model accuracy is positively correlated with the learning rate. When $\alpha \geq 0.005$, as the learning rate increases, the model accuracy decreases slightly. The final choice of learning rate is 0.005.

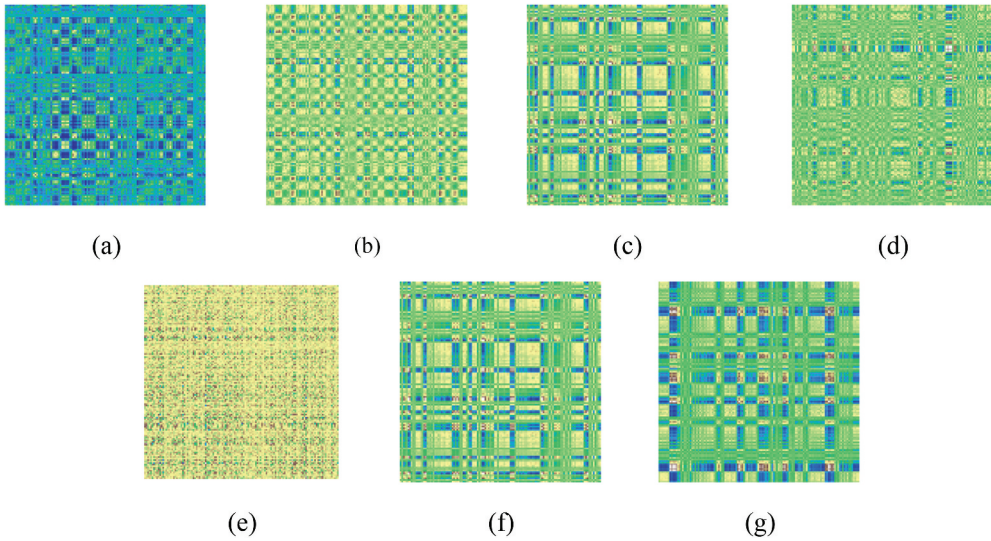


Figure 8. The MTF images of the different bearing health conditions: (a) normal condition, (b) if 0.18, (c) of 0.18, (d) BF 0.18, (e) if 0.36, (f) of 0.36, (g) BF 0.36.

- ③ **Auxiliary classification weight:** In this experiment, the weights of the auxiliary classifier are 0, 0.1, 0.2, 0.3 and 0.4, respectively. It can be seen from Figure 9(c) that the model accuracy is the highest when the weight of the auxiliary classifiers is 0.3. At the same time, it can be seen that the accuracy of the model is not optimal when the auxiliary classification weight is 0 (that is, no auxiliary classifier module is added), which verifies the effectiveness of the auxiliary classifier. Finally, the weight parameter 0.3 is selected.
- ④ **Kernel size:** In this experiment, convolutional kernel sizes 1×1 , 3×3 , 5×5 and 7×7 are designed for the first convolutional layer in the backbone network. From Figure 9(d), it can be seen that the accuracy of the model is positively correlated with the kernel size when the convolutional kernel size is less than 5, and the highest accuracy is achieved when the convolutional kernel is 5×5 .

According to the above hyperparameter influence test, batch size = 12, $\alpha = 0.005$, weight = 0.3, kernel size = 5, epoch = 30, the activation function in each layer convolution is ReLU. The selection of model parameters is shown in Table 3.

4.4. Validity analysis

In the field of machine learning, confusion matrix is commonly used to evaluate the performance of classification model. Therefore, this paper adopts the confusion matrix to visualise the results of MTF-MMCNN model under A-A, B-B and C-C dataset, as shown in Figure 10. The horizontal and vertical coordinates in the figure represent the prediction label and the real label respectively. The values on the diagonal represent the number of samples whose faults are correctly classified.

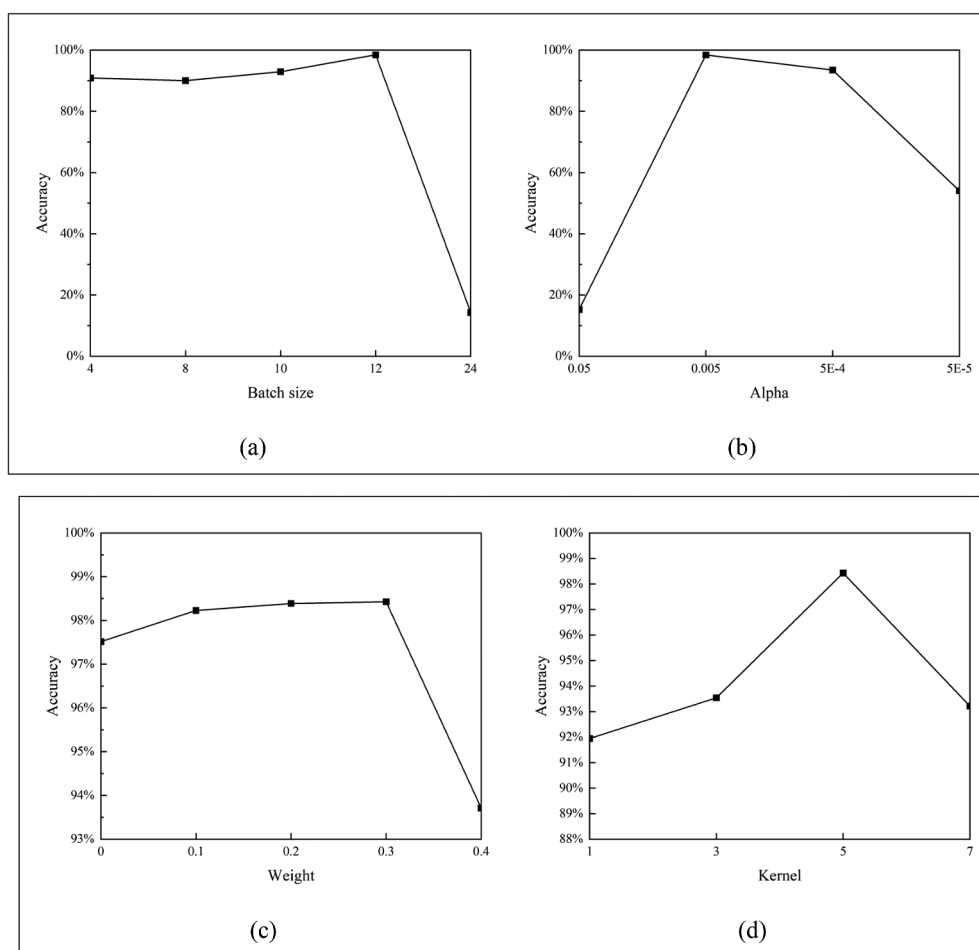


Figure 9. The impact of different parameters on classification accuracy: (a) Batch size, (b) Learning rate, (c) Classification weight, (d) Kernel size.

Taking A-A as an example, there are 100 verification samples in model fault 5, one sample is misclassified into fault 3, and only one of 700 samples is confused in the whole fault sample. The accuracy of the overall validation set reaches 99.86%. At the same time, it can be seen from the diagram that the accuracy of B-B and C-C groups is 99.86% and 96.57%, respectively, indicating that the MTF-MMCNN model has high fault classification accuracy.

5. Comparative analysis experiment

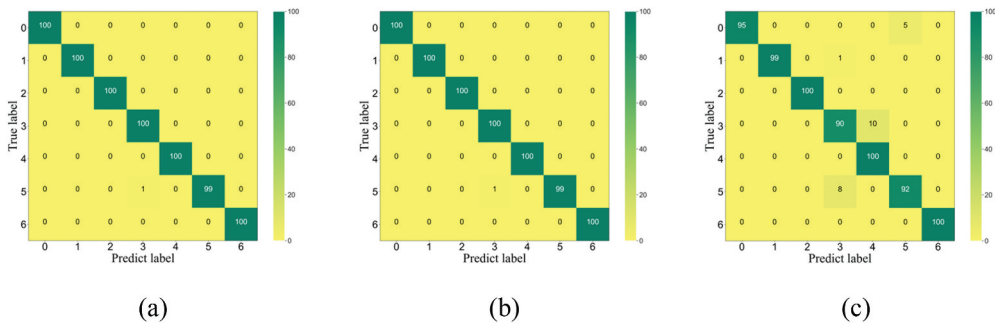
5.1. Diagnostic results under variable load conditions

5.1.1. Comparison of different method

In practical work, the bearing will operate under different load conditions. Therefore, it is necessary to verify the fault diagnosis performance of MTF-MMCNN under variable load

Table 3. Structural parameters of the MDCNN model.

No.	Layer type	Number of filters	Filter size	Output size
1	Input	/	/	$128 \times 128 \times 1$
2	Conv1	16	5×5	$128 \times 128 \times 16$
3	Maxpool1	/	2×2	$64 \times 64 \times 16$
4	BN1	/	/	$64 \times 64 \times 16$
5	Conv2	24	3×3	$64 \times 64 \times 24$
6	Maxpool2	/	2×2	$32 \times 32 \times 24$
7	BN2	/	/	$32 \times 32 \times 24$
8	Branch1	/	/	$32 \times 32 \times 104$
9	Attention1	/	/	$32 \times 32 \times 104$
10	Conv3	104	2×2	$16 \times 16 \times 104$
11	BN3	/	/	$16 \times 16 \times 104$
12	Branch2	/	/	$16 \times 16 \times 160$
13	Attention2	/	/	$16 \times 16 \times 160$
14	Fusion	/	/	$32 \times 32 \times 104$
15	Add	(BN1)	/	$32 \times 32 \times 104$
16	GAP1	(BN2)	/	32
17	Dropout1	/	0.5	32
18	Softmax1	7	/	7
19	GAP2	/	/	32
20	Dropout2	/	0.5	32
21	Softmax2	7	/	7
22	Output	(3:7)	/	7


Figure 10. Confusion matrix of fault prediction results: (a) A-A, (b) B-B, (c) C-C.

conditions. In this paper, three different load datasets A, B, and C shown in Table 1 are combined in pairs and used as training sets and test sets, respectively. In the subsequent experiments, the average value of fault diagnosis accuracy of 10 experiments is taken to observe the results.

In this experiment, the MTF-MMCNN model is compared with Improved LeNet-5 [25], Improved Inception V2 [32], CFNet [33] and MACNN [34]. Among them, LeNet-5 adds more convolutional and pooling layers and BN layers; Inception V2 is a classical multi-scale branch network, which is mainly improved by adding CBAM attention mechanism; CFNet is a multi-scale cascade fusion network for image segmentation; MACNN is a multi-scale neural network that adds ECA attention. The calculation results are shown in Figure 11. Through comparison, it can be seen that the fault diagnosis accuracy of this method is higher than that of other models under variable load conditions. The average accuracy of MTF-MMCNN model is 99.11%, which is 11.99% higher than that of improved LeNet-5, 1.1% higher than Improved InceptionV2, 7.52% higher

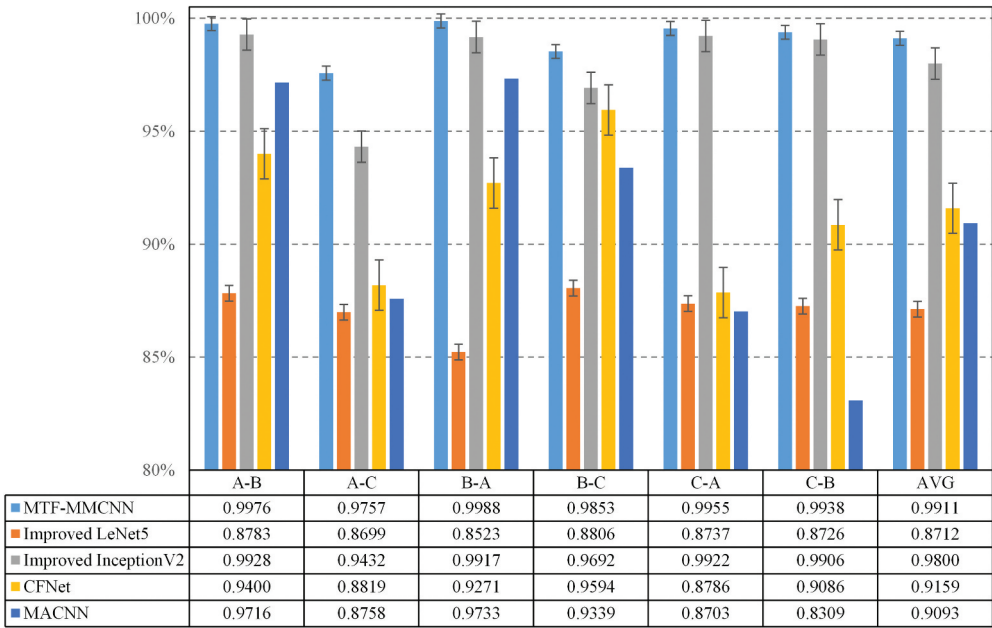


Figure 11. Fault classification accuracy of CWRU dataset under variable load.

Table 4. Experimental results under different dataset size.

Method		MTF-MMCNN	Improved LeNet-5	CFNet	Improved InceptionV2	MACNN
Dataset size		Accuracy (%)				
10%	A-A	98.66	57.54	96.74	80.83	63.05
	B-B	98.72	53.97	97.60	78.23	52.62
	C-C	97.83	61.82	94.27	81.67	55.14
30%	A-A	99.19	77.10	99.17	83.89	74.69
	B-B	99.57	75.59	99.23	89.91	69.84
	C-C	99.37	77.61	99.03	86.40	73.70
60%	A-A	99.77	89.43	99.63	95.49	90.76
	B-B	99.71	86.04	99.57	93.85	92.28
	C-C	99.63	84.23	99.43	91.83	90.52
100%	A-A	99.97	95.23	99.67	95.68	96.52
	B-B	100	95.11	99.86	95.52	98.43
	C-C	99.88	94.88	99.57	95.71	95.24

than CFNet, and 8.18% higher than MACNN model. At the same time, it can be found that the model has high stability by observing the error line. Experiments show that MTF-MMCNN still has high fault diagnosis accuracy under variable load conditions.

5.1.2. Comparison of different datasets scale

In this section, the CWRU dataset is used to verify the accuracy of the MTF-MMCNN model under different dataset sizes. The experimental results are shown in Table 4. It can be seen from Table 4 that the fault classification accuracy of MTF-MMCNN is higher than that of other models under four datasets. Among them, the improved LeNet-5, the improved Inception V2 network and MACNN have low accuracy under the condition of 10% sample size. The CFNet model shows high accuracy of sufficient samples, low

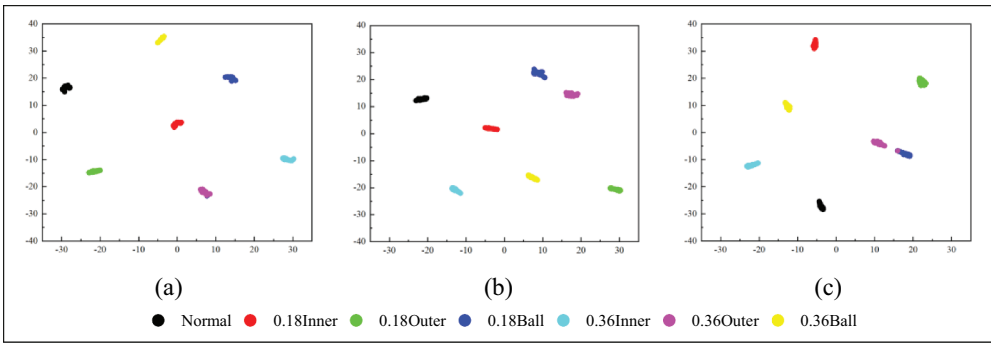


Figure 12. T-sne visualisation under 10% sample size: (a) A-A, (b) B-B, (c) C-C.

accuracy of few samples and poor stability. The accuracy of the MTF-MMCNN model is more than 99.88% in the full sample. As the number of samples decreases, the accuracy of MTF-MMCNN fault classification decreases slightly, moreover, the fluctuation range is the smallest in the experiment. Therefore, the MTF-MMCNN model also has better fault classification ability and robustness when the number of fault samples is small.

In order to clearly observe the fault classification accuracy of MTF-MMCNN under different sample sizes, the t-SNE (t-distributed Stochastic Neighbor Embedding) is applied to reduce the high-dimensional data to two-dimension, and the results are drawn into a scatter plot, as shown in Figure 12. The seven fault labels of the bearing are represented by different colours. It can be seen from Figure 12(a) that MTF-MMCNN can still clearly classify the faults into different categories even when the sample size is reduced to 10%. No sample aliasing is observed in Figure 12(b). In Figure 12(c), a 0.36 mm outer ring fault sample is superimposed with a 0.18 mm rolling element fault sample. Therefore, the model can achieve better fault aggregation classification, indicating that the model has strong fault diagnosis stability.

5.2. Diagnostic results under variable speed conditions

5.2.1. Comparison of different method

The generalisation performance test results of MTF-MMCNN model under variable speed conditions are shown in Figure 13. Taking the experimental D-E as an example, the accuracy of the model reaches 99.26%, which is 10.52% higher than the Improved LeNet-5, 24.32% higher than the Improved Inception V2, 2.80% higher than CFNet, and 17.64% higher than MACNN. It can be seen from the error line that the model is more stable than other algorithms. It shows that the model proposed in this paper has stable accuracy and is not easily disturbed by external factors. The experiment verifies that the MTF-MMCNN model has high stability under variable speed conditions, and can deal with the situation with variable speed in practical applications.

5.2.2. Ablation experiment

In this section, ablation experiments are designed to verify the effectiveness of the proposed modules. Each experiment uses 30 training samples, and the results are

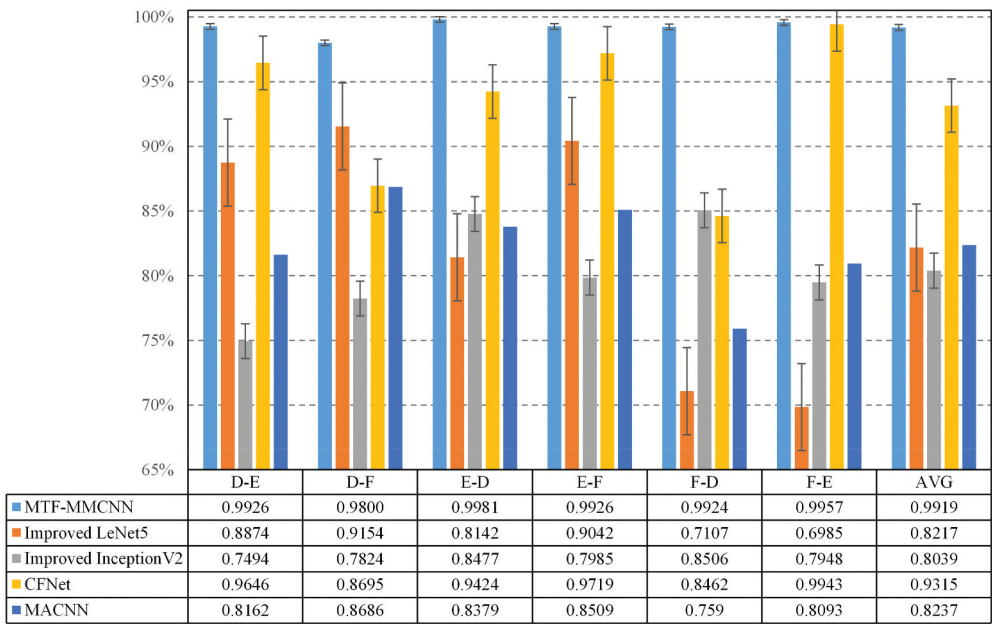


Figure 13. Fault classification accuracy of MFS dataset under variable speed condition.

Table 5. Ablation experimental of MFS dataset.

Experiment	D-D	E-E	F-F	AVG
Method	Accuracy (%)			
MTF-MMCNN+CFAM	95.62	98.41	96.82	96.95
1D-MMCNN+CFAM	90.43	90.19	87.71	89.44
MTF-MMCNN	83.14	84.82	87.81	85.26
MTF-MMCNN+CBAM	81.05	82.05	88.66	83.92
MTF-MB+CFAM	71.52	61.09	74.66	69.09
MTF-MC+CFAM	70.81	63.19	47.62	60.54

shown in Table 5. Among them, MTF represents the use of Markov transfer field to generate two-dimensional image. 1D represents the direct feature extraction of the original one-dimensional data. MMCNN is a mixed multi-scale feature extraction network model designed in this paper. CFAM is the convolutional fusion attention mechanism proposed in this paper. CBAM is a dual attention module. MB indicates that the feature extraction network only contains multi-scale branch modules. The MC represents that the feature extraction network contains only multi-scale cascade modules.

It can be seen from Table 5 that the MTF-MMCNN+CFAM model has higher average prediction accuracy than the other five structures under the same speed condition. Taking the experimental D-D as an example, the MTF coding method is 5.19% higher than the 1D convolution accuracy. The accuracy of using CFAM module is 12.48% higher than that of without channel attention and 14.57% higher than that of using CBAM module. MMCNN is 24.10% higher than that using only branch module and 24.81% higher than that using only cascade module. It can be clearly seen that the mixed multi-scale network can effectively

improve the accuracy of fault diagnosis. This experiment supports the conclusion that the two-dimensional image converted by MTF method can retain more original data information. The introduction of CFAM can also reasonably allocate network channel weights. At the same time, in the parameter sensitivity experiment, it can also be found that the auxiliary classifier has a certain effect on the improvement of model accuracy. Therefore, the MTF-MMCNN model and CFAM module proposed in this paper can effectively improve the accuracy of fault classification.

6. Conclusion

In order to solve the problem of low fault classification accuracy of rolling bearings under variable working conditions, this paper proposes a rolling bearing fault diagnosis method based on MTF and MMCNN. The method is verified by the CWRU dataset and the MFS dataset of our laboratory and compared with other algorithms. The following conclusions are drawn:

- (1) The MMCNN module is proposed in this paper, which integrates multi-scale branches and multi-scale cascaded structures, and can utilise different scale convolution kernels to obtain different scale information, fuse shallow spatial information and deep semantic information to extract more features from vibration signal.
- (2) The CFAM channel attention module presented can adaptively assign the weight of each channel during the training process of the network model, and pinpoint valuable features more precisely and extract more potent feature information with a few samples.
- (3) After the shallow network feature screening, the auxiliary classifier is connected and the fault classification is achieved with the main classifier according to their respective weights, which effectively prevents the feature loss caused by the deepening of the network layer and improves the fault recognition performance of the model in the changing environment of working conditions.

Although the model proposed has excellent performance, there is still potential for further development. In order to consider the submerged influence of noise on the original vibration signal in practical work, future work should focus on enhancing the fault diagnosis ability of the model under strong noise environments to adapt to a wider range of scenarios.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the National Natural Science Foundation of China [No. 52465013, No. 51465035] the Natural Science Foundation of Gansu Province [No. 20JR5RA466].

ORCID

Qiyue Zhang  <http://orcid.org/0009-0001-1850-3664>

Data availability statement

The data cannot be made publicly available upon publication because they contain sensitive personal information. The data that support the findings of this study are available upon reasonable request from the authors.

References

- [1] Lei ZH, Zhang P, Chen YJ, et al. Prior knowledge-embedded meta-transfer learning for few-shot fault diagnosis under variable operating conditions. *Mech Syst Signal Process.* 2023;200:110491. doi: [10.1016/j.ymssp.2023.110491](https://doi.org/10.1016/j.ymssp.2023.110491)
- [2] Qiao ZG, Yao DC, Yang JW, et al. MSTD: a framework for rolling bearing fault diagnosis based on multi-scale and soft-threshold denoising. *Nondestr Test Eval.* 2024;1–21. doi: [10.1080/10589759.2024.2430384](https://doi.org/10.1080/10589759.2024.2430384)
- [3] Zhang SH, Liang WH, Zhao WZ, et al. Electro-hydraulic SBW fault diagnosis method based on novel 1DCNN-LSTM with attention mechanisms and transfer learning. *Mech Syst Signal Process.* 2024;220:111644. doi: [10.1016/j.ymssp.2024.111644](https://doi.org/10.1016/j.ymssp.2024.111644)
- [4] Liu T, Li LX, Noman K, et al. Sliding time synchronous averaging based on independent extended autocorrelation function for feature extraction of bearing fault. *Measurement.* 2024;236:115130. doi: [10.1016/j.measurement.2024.115130](https://doi.org/10.1016/j.measurement.2024.115130)
- [5] Long Z, Guo JY, Ma XG, et al. Motor fault diagnosis based on multisensor-driven visual information fusion. *ISA Trans.* 2024;155:524–535. doi: [10.1016/j.isatra.2024.09.024](https://doi.org/10.1016/j.isatra.2024.09.024)
- [6] Lei CL, Miao CX, Yu YH, et al. Rolling bearing fault diagnosis method based on multi-scale pooling residual convolutional neural network under noisy environment. *Eksploracja i Niezawodność – Maint Reliab.* 2025;27(1). doi: [10.17531/ein/192167](https://doi.org/10.17531/ein/192167)
- [7] Waziralilah NF, Abu A, Lim MH, et al. A review on convolutional neural network in bearing fault diagnosis[c]. In: *MATEC Web of Conferences. Engineering Application of Artificial Intelligence Conference 2018; Sabah, Malaysia; 2019 Dec; Vol255.* p. 06002.
- [8] Liu T, Li LX, Khandaker N, et al. Local maximum instantaneous extraction transform based on extended autocorrelation function for bearing fault diagnosis. *Adv Eng Inf.* 2024;61:102487. doi: [10.1016/j.aei.2024.102487](https://doi.org/10.1016/j.aei.2024.102487)
- [9] Saidi L, Ali JB, Fnaiech F. Bi-spectrum based-emd applied to the non-stationary vibration signals for bearing faults diagnosis. *ISA Trans.* 2014;53(5):1650–1660. doi: [10.1016/j.isatra.2014.06.002](https://doi.org/10.1016/j.isatra.2014.06.002)
- [10] Lei CL, Xia BF, Xue LL, et al. Rolling bearing fault diagnosis method based on MTF-CNN. *J Vibr Shock.* 2022;41(9):151–158.
- [11] Hinton GE, Salakhutdinov RR. Reducing the dimensionality of data with neural networks. *Science.* 2006;313(5786):504–507. doi: [10.1126/science.1127647](https://doi.org/10.1126/science.1127647)
- [12] Jiang LL, Li SH, Li XJ, et al. Fault diagnosis of a planetary gearbox based on a local bi-spectrum and a convolutional neural network. *Meas Sci Technol.* 2022;33(4):045008. doi: [10.1088/1361-6501/ac471a](https://doi.org/10.1088/1361-6501/ac471a)
- [13] Pareek V, Chaudhury S. Deep learning-based gas identification and quantification with auto-tuning of hyper-parameters. *Soft Comput.* 2021;25(22):14155–14170. doi: [10.1007/s00500-021-06222-1](https://doi.org/10.1007/s00500-021-06222-1)
- [14] Yang B, Wang W, Chen GQ, et al. Improving image-based weld defects detection under interference information via a novel semantic supervision strategy. *Nondestr Test Eval.* 2024;1–17. doi: [10.1080/10589759.2024.2397706](https://doi.org/10.1080/10589759.2024.2397706)
- [15] Liu C. Robotic emotion recognition using two-level features fusion in audio signals of speech. *IEEE Sensors J.* 2021;22(18):17447–17454. doi: [10.1109/JSEN.2021.3065012](https://doi.org/10.1109/JSEN.2021.3065012)

- [16] Li S, Ji JC, Feng K, et al. Composite neuro-fuzzy system-guided cross-modal zero-sample diagnostic framework using multi-source heterogeneous non-contact sensing data. *IEEE Trans Fuzzy Syst.* **2025**;33(1):302–313. doi: [10.1109/TFUZZ.2024.3470960](https://doi.org/10.1109/TFUZZ.2024.3470960)
- [17] Lei CL, Wang L, Zhang QY, et al. Rolling bearing fault diagnosis method based on MTF and PC-MDCNN. *J Mech Sci Technol.* **2024**;38(7):3315–3325. doi: [10.1007/s12206-024-0606-y](https://doi.org/10.1007/s12206-024-0606-y)
- [18] Feng K, Ji JC, Yang YC, et al. Digital twin-driven intelligent assessment of gear surface degradation. *Mech Syst Signal Process.* **2023**;186:109896. doi: [10.1016/j.ymssp.2022.109896](https://doi.org/10.1016/j.ymssp.2022.109896)
- [19] Xu YD, Ji J, Ni Q, et al. A graph-guided collaborative convolutional neural network for fault diagnosis of electromechanical systems. *Mech Syst Signal Process.* **2023**;200:110609. doi: [10.1016/j.ymssp.2023.110609](https://doi.org/10.1016/j.ymssp.2023.110609)
- [20] Feng K, Ni Q, Chen Y, et al. A cyclostationarity-based wear monitoring framework of spur gears in intelligent manufacturing systems. *Struct Health Monit.* **2023**;22(5):3092–3108. doi: [10.1177/14759217221147018](https://doi.org/10.1177/14759217221147018)
- [21] Feng HN, Fu S, Xu YG. Fault diagnosis of rotating machinery based on BN-1DCNN model. *J Vibr Shock.* **2021**;40(19):302–308.
- [22] Liu ZL, Wang H, Liu JJ, et al. Multitask learning based on lightweight 1DCNN for fault diagnosis of wheelset bearings. *IEEE Trans Instrum Meas.* **2020**;70:1–11. doi: [10.1109/TIM.2020.3017900](https://doi.org/10.1109/TIM.2020.3017900)
- [23] Huang WY, Cheng JS, Yang YY, et al. An improved deep convolutional neural network with multi-scale information for bearing fault diagnosis. *Neurocomputing.* **2019**;359:77–92. doi: [10.1016/j.neucom.2019.05.052](https://doi.org/10.1016/j.neucom.2019.05.052)
- [24] Gu YK, Chen RH, Huang P, et al. A lightweight bearing compound fault diagnosis method with gram angle field and ghost-ResNet model. *IEEE Trans Rel.* **2024**;73(4):1768–1781. doi: [10.1109/TR.2023.3332223](https://doi.org/10.1109/TR.2023.3332223)
- [25] Wu CF, Yang SX, Huang HZ, et al. Research on fault diagnosis method of rolling bearing based on improved LeNet-5 model. *J Vibr Shock.* **2021**;40(12):55–61.
- [26] Lei CL, Xue LL, Jiao MX, et al. Rolling bearing fault diagnosis by Markov transition field and multi-dimension convolutional neural network. *Meas Sci Technol.* **2022**;33(11):114009. doi: [10.1088/1361-6501/ac87c4](https://doi.org/10.1088/1361-6501/ac87c4)
- [27] Wang H, Liu ZL, Peng DD, et al. Understanding and learning discriminant features based on multiattention 1DCNN for wheelset bearing fault diagnosis. *IEEE Trans Ind Inf.* **2019**;16(9):5735–5745. doi: [10.1109/TII.2019.2955540](https://doi.org/10.1109/TII.2019.2955540)
- [28] Hu J, Shen L, Sun G. Squeeze-and-excitation networks[c]. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*; Salt Lake City, (UT), USA; **2018** Jun; p. 7132–7141.
- [29] Wang H, Xu JW, Yan RQ, et al. Intelligent bearing fault diagnosis using multi-head attention-based CNN. *Procedia Manuf.* **2020**;49:112–118. doi: [10.1016/j.promfg.2020.07.005](https://doi.org/10.1016/j.promfg.2020.07.005)
- [30] Xu SJ, Yuan R, Lv Y, et al. A novel fault diagnosis approach of rolling bearing using intrinsic feature extraction and cbam-enhanced InceptionNet. *Meas Sci Technol.* **2023**;34(10):105111. doi: [10.1088/1361-6501/ace19c](https://doi.org/10.1088/1361-6501/ace19c)
- [31] Case Western Reserve University Bearing Data Center [EB/OL]. **2018**. Available from: <https://csegroups.case.edu/bearingdatacenter/pages/download-data-file>
- [32] Yao QS, Bie SS, Yu JH, et al. A bearing fault diagnosis method combining improved Inception V2 module and CBAM is proposed. *J Vibr Eng.* **2022**;35(4):949–957.
- [33] Zhang G, Li ZY, Tang JMCF, et al. CEDNet: A cascade encoder–decoder network for dense prediction. *Pattern Recognition.* **2025** 158 111072.
- [34] Zhang XJ, Shang JY, Yu GJ, et al. Attention-based multi-scale convolutional neural network bearing fault diagnosis. *J Jilin Univ(eng Technol Ed).* **2024**;54 10 3009–3017.